

Institute of Actuaries of India

Subject CM1A – Actuarial Mathematics (Paper A)

September 2021 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1: Option A

Workings for reference (no marks to be awarded for showing steps as this is an MCQ)

$$i = 9\%$$

$$\text{So, } i^{(2)} = 0.0880613$$

$$a_{10\overline{}}^{(2)} = 6.558944$$

$$v^{10} = 0.42241$$

$$\text{So PV} = 7 \times a_{10\overline{}}^{(2)} + 100 \times v^{10}$$

$$= 88.15$$

[4 Marks]**Solution 2:** The risks covered by car insurance include:

- The insurer will provide cover to pay for damage to the insured vehicle or fire or theft of the vehicle, known as property cover.
- The contract also provides cover for compensation payable to third parties for death, injury or damage to their property, known as liability cover. (2)

From the perspective of the policyholder, the cash flows are:

- Negative cashflow from the premium paid either as a single lumpsum at the beginning or at monthly intervals. (0.5)
- Positive cashflow when the insurance company settles the claim. (0.5)
- If there are delays in the insurance company settling the claim, there could be a further negative cashflow from the policyholder to pay for the repairs or compensation before the insurance company reimburses for the claim. (1)
- If the insurance company directly settles the claim with the repair company or the third party, the policyholder may not experience further cashflows (1)

[5 Marks]**Solution 3:**

- Present value of an immediate annuity of 1 unit per unit time interval / per annum paid in arrear for as long as the status n_1 remains active/ as long as the term n does not expire/ over a period of length n . (1.5)
- Expected Present Value of an immediate annuity of 1 unit per annum paid in arrear for as long as the status 'x' remains alive in future/person remains alive. (1.5)
- Principle of equivalence / equation of value:
 Expected present value of income = Expected present value of outgo (0.5)
 $E(\text{present value of the outgo}) - E(\text{present value of the Income}) = 0$ (0.5)
 $\Rightarrow E(\text{Present value of future loss}) = 0$ (0.5)
 i.e. $E(L_0) = 0$ (0.5)

[5 Marks]**Solution 4:** Option A

Workings for reference (no marks to be awarded for showing steps as this is an MCQ)

The probability of staying with the company for each year in future = $1 - 7\% = 0.93$

The total probability of continuing with the company for 25 years is $(0.93)^{25} = 0.16295$

Therefore, EPV of the benefit is equal to:

$$1000000 \times \frac{1}{1.05^{25}} \times 0.93^{25} = 48,122$$

$${}^2E(PV) = E(PV) @ 10.25\% = \left(\frac{1}{1.1025^{25}}\right) * 0.93^{25} = 0.014211$$

$$\text{Variance} = 1000,000^2 * ({}^2E(PV) - (E(PV))^2 = 11,894,779,461$$

$$Sd(PV) = \sqrt{\text{Variance}}$$

$$= 109,063$$

[4 Marks]

Solution 5: Option D

Workings for reference

$${}_{10}p_{50} = l_{60} / l_{50} = 0.956255$$

$$A_{60} = 0.32692$$

$$v^{10} = 1.06^{-10} = 0.5583948$$

$$\text{Benefit} = 200,000$$

$$\text{Present value of benefit} = 0.956255 \times 0.32692 \times 0.5583948 \times 200,000 = 34,912$$

[4 Marks]

Solution 6:

i) We have $a_x = \sum_{j=1}^{\infty} j p_x v^j$ (1)

Where $j p_x = e^{-0.03j}$ and $v^j = e^{-\delta j} = e^{-0.05j}$ (1)

So $a_x = \sum_{j=1}^{\infty} e^{-0.05j} * e^{-0.03j} = \sum_{j=1}^{\infty} e^{-0.08j}$ (0.5)

This is sum to the infinity of a geometric progression with first term $a = e^{-0.08}$ and common ratio $r = e^{-0.08}$.

So using the formula:

$$S_{\infty} = \frac{a}{1-r} \text{ gives,}$$

$$a_x = \frac{e^{-0.08}}{1 - e^{-0.08}} = 12.0067$$
 (1.5)

[4]

ii) $\ddot{a}_x = a_x + 1 = 12.0067 + 1 = 13.0067$ (1)

[5 Marks]

Solution 7:

The cash flows expected to occur in the 30th month are:

- Premium income of INR 5000 provided policyholder is alive at the beginning of the month
- Renewal expense outgo, provided policyholder is alive at the beginning of the month.
- Death benefit outgo based on reduced sum assured of the contract, provided the policyholder dies in the 30th month.
- Renewal commission outgo based on the premium of the contract, provided the policyholder is alive at the beginning of the 30th month.

Reduced sum assured at the start of the 30th month is = $5,000,000 - 20,000 \ddot{s}_{30}^{@1\%}$ (1)
 $= 4297345$ (0.5)

Expense outgo amount for 30th month = $\text{INR } 50 * (1.03^{28/12}) = \text{INR } 53.57$ (1)

The probability that the policyholder survives to the beginning of the 24th month is:

$\frac{l_{57}}{l_{55}}$, where all mortality rates considered allow for the 3-year set back in AM92 table

i.e. for l_{57} , we seek value of l_{54} from AM92 table (and similarly for all other ages)

Assuming deaths are uniformly distributed over the year of age, we can calculate the value of l_x at the non - integer age using linear interpolation between the values at adjacent integer ages. So, the required probability is equal to:

$$\frac{(\frac{6}{12}l_{57} + \frac{6}{12}l_{58})}{l_{55}} = \frac{(\frac{6}{12} * 9595.9715 + \frac{6}{12} * 9557.8179)}{9660.5021} = 0.991345 \quad (1)$$

The probability that the policyholder dies in the 30th month is:

$$\frac{1}{12} * \frac{d_{57}}{l_{55}} = \frac{38.1536}{12 * 9660.5021} = 0.000329 \quad (0.5)$$

Expected premium income = 5000 x 0.991345 = 4956.7 (0.5)

Expected expense outgo = 53.57 x 0.991345 = 53.1 (0.5)

Expected death outgo = 4297345 x 0.000329 = 1414.3 (0.5)

Expected renewal commission outgo = 5% x 4956.7 = 247.8 (0.5)

So, net cash flow = income – outgo = premium – expenses – death – commission
= 3,241.4

(1)

[7 Marks]

Solution 8:

- i) $\mu_d = \text{force of death} = 0.01$
 $\mu_s = \text{force of sickness} = 0.04$
 $\mu_{lap} = \text{force of lapse} = 0.05$

Probability of surviving the duration of 1 year without exit:

$$(ap) = e^{-(0.1)} = 0.90483742 \quad (1)$$

Dependent rates can be calculated as:

$$\begin{aligned} (aq)^d &= \frac{\mu^d}{\mu^d + \mu^s + \mu^{lap}} (1 - e^{-(\mu^d + \mu^s + \mu^{lap})}) = \frac{0.01}{0.1} (1 - e^{-(0.1)}) \\ &= 0.0095163 \\ (aq)^s &= \frac{\mu^s}{\mu^d + \mu^s + \mu^{lap}} (1 - e^{-(\mu^d + \mu^s + \mu^{lap})}) = \frac{0.04}{0.1} (1 - e^{-(0.1)}) = 0.038065 \\ (aq)^{lap} &= \frac{\mu^{lap}}{\mu^d + \mu^s + \mu^{lap}} (1 - e^{-(\mu^d + \mu^s + \mu^{lap})}) = \frac{0.05}{0.1} (1 - e^{-(0.1)}) = 0.047581 \end{aligned} \quad (3)$$

[4]

- ii) PV Outgo due to payment of Survival Benefit

$$= 300,000 \times 0.90483742^5 \times (1.05)^{-5} = 142,570 \quad (1)$$

PV Outgo due to death

$$\begin{aligned} &= 800,000 \times ((aq)^d v^1 + (ap) (aq)^d v^2 + (ap)^2 (aq)^d v^3 + (ap)^3 (aq)^d v^4 + \\ &(ap)^4 (aq)^d v^5) \\ &= 800000 \times 0.0344016 = 27,521 \end{aligned} \quad (1.5)$$

PV Outgo due to sickness

$$\begin{aligned} &= 500000 \times ((aq)^s v^1 + (ap) (aq)^s v^2 + (ap)^2 (aq)^s v^3 + (ap)^3 (aq)^s v^4 + \\ &(ap)^4 (aq)^s v^5) \\ &= 500000 \times 0.137606 = 68,803 \end{aligned} \quad (1.5)$$

PV Outgo due to lapsation

$$= 60\% \times 100,000 \times ((aq)^{lap} v^1 + 2 \times (ap) (aq)^{lap} v^2 + 3 \times (ap)^2 (aq)^{lap} v^3 + 4 \times (ap)^3 (aq)^{lap} v^4 + 5 \times (ap)^4 (aq)^{lap} v^5)$$

=)

$$= 60000 \times 0.465322 = 27,919 \quad (1.5)$$

PV of Premium

$$= 100,000 \times (1 + (ap) \times v^1 + (ap)^2 \times v^2 + (ap)^3 \times v^3 + (ap)^4 \times v^4)$$

$$= 100,000 \times 3.795783$$

$$= 379,578 \quad (1.5)$$

$$\text{Net Present Value} = 379,578 - 68,803 - 27,521 - 27,919 - 142,570 = 112,765 \quad (1)$$

[8]

[12 Marks]

Solution 9:

Option B

[3 Marks]

Working for reference:

$$\text{Percentage in reserves} = 1360 / 1500 - 1 = -9.3\%$$

$$\text{Change in interest rate} = 10\% \times 6\% = 0.06\%$$

$$\text{Duration} = 9.3\% / 0.06\% = 15.66 \text{ year} \sim 16 \text{ years}$$

Solution 10:

i) Option B (3)

Workings for reference (no marks to be awarded for showing steps as this is an MCQ)

Number of days the bill is held by the first purchaser = t

$$103 \times (1.05)^t = 104$$

$$t = \log 1.009709 / \log 1.05 = 0.198$$

Converted to days, this is $0.198 \times 365 = 72.2809$ or 72 days

ii) Option B (3)

Workings for reference (no marks to be awarded for showing steps as this is an MCQ)

Number of days the bill was held by second investor = $91 - 72 = 19$ days

$$104 \times (1 + \frac{19}{365} i) = 105$$

Solving this, we get $i = 0.1875 = 19\%$ return

[6 Marks]

Solution 11:

Let the monthly premium be P.

The annual premium = 12P

The benefits payable under the product are:

$$\text{Value of lumpsum benefit} = 500,000 \bar{A}_{60:60}$$

$$= 500,000 \times (1 + 0.04)^{0.5} A_{60:60} \quad (0.5)$$

$$A_{60:60} = 1 - d\ddot{a}_{60|60} = 1 - 0.038462 \times 14.090 = 0.45807 \quad (0.5)$$

$$\text{Therefore, value of benefit} = 233,571 \quad (0.5)$$

Annuity payable to female upon death of the male life

$$400,000 v p_{60} \ddot{a}_{60|61}^{(m)} \quad (1)$$

$$= 400,000 v p_{60} (\ddot{a}_{61}(f) - \ddot{a}_{60:61}^{(m)})$$

$$= \frac{400,000}{1.04} * (1 - 0.002058) * (16.311 - 13.918) \quad (0.5)$$

$$= 918,491 \quad (0.5)$$

Annuity payable to male upon death of the female life

$$+ + 200,000 v p_{60} \ddot{a}_{60|61}^{(f)} \quad (1)$$

$$= 200,000 v p_{60} \left(\ddot{a}_{61}(m) - \ddot{a}_{60:61}^{(f)} \right)$$

$$= 200,000 \frac{1}{1.04} * (1 - 0.002451) * (15.254 - 13.837) \quad (0.5)$$

$$= 271832 \quad (0.5)$$

$$\text{Value of benefits} = 233571 + 271832 + 918491 = 1,423,894$$

Value of premium =

$$12 x P x \ddot{a}^{(12)}_{60:60:20} =$$

$$12P x \left(\ddot{a}_{60:60} - \frac{11}{24} - v^{20} {}_{20}p_{60} {}_{20}p_{60} \left(\ddot{a}_{80:80} - \frac{11}{24} \right) \right) \quad (0.5)$$

$$= 12P x \left(14.09 - 11/24 - (1.04)^{-20} x \left(\frac{6953.536}{9826.131} \right) \left(\frac{7724.737}{9848.431} \right) x \left(5.857 - \frac{11}{24} \right) \right) \quad (1)$$

$$= 147.2P$$

$$\text{Initial commission} = 20\% * 12P = 2.4P \quad (0.5)$$

$$\text{Renewal commission and expenses} = 3\% * 147.2 * P = 4.416P \quad (0.5)$$

$$\text{Initial expense} = 10\% * 12P = 1.2P \quad (0.5)$$

So, from principle of equivalence we have:

$$147.2P = 1,423,894 + 2.4P + 4.416P + 1.2P \quad (0.5)$$

$$139.184P = 1,423,894 \quad (0.5)$$

$$P = 10,230$$

[10 Marks]

Solution 12:

i) Monthly effective interest rate = $i^{(12)}/12 = (1+i)^{(1/12)} - 1 = 0.5262\%$ (1)

Annuity function (in arrears) $a_{60}^{@0.5262\%} = 51.33692$ (1)

X = PV of installments on 1 March 2017 = $200,000 * a_{60}^{@0.5262\%}$
 = INR 10,267,383 (1)

Loan amount as at March 2015 = $X/(1+6.5\%)^2 = \text{INR } 9,052,334$

Savings consumed = $15,000,000 - 9,052,334 = 5,947,666$

(1)

[4]

ii) As on 1 March 2020, there are 24 monthly instalments due for the student loan

Annuity function (in arrears) $a_{24}|^{0.5262\%} = 22.49$

(1)

Outstanding loan amount as at 1 March 2020 = $2,00,000 \times 22.49 = \text{INR } 4,498,198$

(1)

New loan amount = $14,498,198$

(1)

This is to be paid in 10 annual instalments in arrears at 5% p.a.

$a_{10}|^{5\%} = 7.72$

(1)

Annual instalment = $14,498,198 / 7.72 = 1,877,583$

(1)

[5]

iii) Annuity function in advance $a(\text{due})_{12}|^{5\%} = 9.31$

(1)

Loan amount = $\text{INR } 10,000,000$

Instalment = $\text{INR } 10,000,000/9.31 = 1,074,528$

(1)

Total loan amount as on 31 March 2020 = $14,498,198$

Choice A = 10 instalments of $\text{INR } 1,877,583$

So rupee value of interest payment = $1,877,583 \times 10 - 14,498,198$

$= 4,277,632$

(1)

Choice B = 24 instalments of $\text{INR } 2,00,000$ and 12 instalments of $\text{INR } 10,74,528$

So rupee value of interest payment = $2,00,000 \times 24 + 10,74,528 \times 12 - 14,498,198$

$= 1,76,94,333 - 14,498,198$

$= 31,96,134$

(1)

Choice B has lower cumulative interest – so Pallavi will choose Option B

(1)

[5]**[14 Marks]**

Solution 13: Let P be the monthly premium

Present value of premiums = present value of decreasing benefits + present value of annuity benefit + present value of initial expenses + present value of regular expenses + present value of claim expenses

(0.5)

Present value of premium = $12Pa_{[30]:30}^{(12)} =$

(0.5)

$$a_{[30]:30}^{(12)} \approx \ddot{a}_{[30]:30} - \frac{11}{24} \left(1 - \frac{D_{60}}{D_{[30]}} \right)$$

(0.5)

$$\frac{D_{60}}{D_{[30]}} = \frac{882.85}{3059.68} = 0.288543$$

(0.5)

$$a_{[30]:30}^{(12)} = 17.759 - \frac{11}{24} * (1 - 0.288543) = 17.433$$

(0.5)

Present value of premiums = $209.195 P$

Present value of decreasing benefits = $515000\bar{A}_{[30]:30}^1 - 15000(I\bar{A})_{[30]:30}^1$

(1)

$$\bar{A}_{[30]:30}^1 \approx 1.04^{0.5} * \left(A_{[30]:30} - \frac{D_{60}}{D_{[30]}} \right)$$

$$= 1.04^{0.5} * (0.31697 - .288543) = 0.02899$$

(1)

$$\begin{aligned}
 (IA)_{[30]:30}^1 &\approx 1.04^{0.5} \left((IA)_{[30]} - \frac{D_{60}}{D_{[30]}} ((IA)_{60} + 30A_{60}) \right) \\
 &= 1.04^{0.5} (6.91644 - 0.288543(8.36234 + 30 * 0.4564)) = 0.56376
 \end{aligned}
 \tag{1}$$

$$\begin{aligned}
 \text{Present value of decreasing benefits} &= 515000 \times 0.02899 - 15000 \times 0.56376 \\
 &= 6473.45
 \end{aligned}
 \tag{0.5}$$

$$\text{Present value of annuity benefit} = 20000 * \frac{D_{60}}{D_{[30]}} a_{60}^{(12)}
 \tag{1}$$

$$a_{60}^{(12)} = \ddot{a}_{60}^{(12)} - \frac{1}{12} \approx \ddot{a}_{60} - \frac{11}{24} - \frac{1}{12} = 15.632 - \frac{13}{24} = 15.09033
 \tag{1}$$

$$\text{Present value of annuity benefit} = 87,084$$

$$\begin{aligned}
 \text{Present value of initial expense} &= 850 \\
 &
 \end{aligned}
 \tag{0.5}$$

$$\begin{aligned}
 \text{Present value of regular expense} &= 50 \ddot{a}_{[30]:30@0\%} \\
 &
 \end{aligned}
 \tag{1}$$

$$\ddot{a}_{[30]:30@0\%} = \ddot{a}_{[30]@0\%} - (v@0\%)^{30} \frac{l_{60}}{l_{[30]}} \ddot{a}_{60@0\%}
 \tag{0.5}$$

$$\text{Now } v@0\% = 1$$

$$\text{And } \ddot{a}_{[30]@0\%} = 1 + e_{[30]}$$

$$\begin{aligned}
 \ddot{a}_{[30]:30@0\%} &= 1 + e_{[30]} - \frac{l_{60}}{l_{[30]}} (1 + e_{60}) \\
 &= 1 + 48.764 - \frac{9287.2164}{9923.7497} (1 + 20.67) = 29.484
 \end{aligned}
 \tag{1}$$

$$\begin{aligned}
 \text{So, present value of regular expenses} &= 50 \times 29.484 = 1474.2 \\
 &
 \end{aligned}
 \tag{0.5}$$

$$\begin{aligned}
 \text{Present value of claim expenses} &= 0.75\% \times \text{present value of annuity benefit} = 653.12 \\
 &
 \end{aligned}
 \tag{0.5}$$

So, from equation of value we get:

$$209.195 P = 6473.45 + 87084 + 850 + 1474.2 + 653.12
 \tag{0.5}$$

$$P = 461.46
 \tag{0.5}$$

[13 Marks]

Solution 14:

- i) B (2)
- ii) B (2)
- iii) A (2)
- iv) C (2)

[8 Marks]
